

ELECTROMAGNETIC INDUCTION

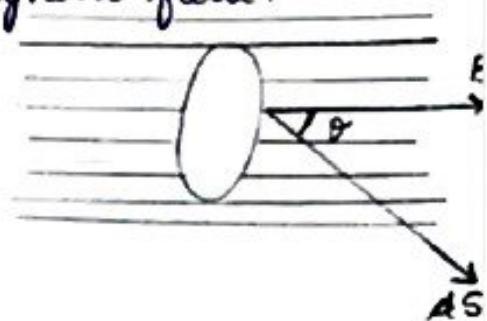
(1)

MAGNETIC FLUX -

It represents total magnetic lines of force passing normally through a given area placed in a magnetic field.

$$\phi_B = B \cdot S = BS \cos \theta$$

Unit - Weber (Wb) = $Tm^2 \rightarrow$ SI unit
Maxwell $\rightarrow$ CGS unit



ELECTROMAGNETIC INDUCTION -

The phenomenon to generate induced current or induced emf by changing the magnetic flux linked with a closed circuit is known as Electromagnetic Induction.

FARADAY'S LAWS

① First Law - Whenever there is change in magnetic flux linked with the closed loop, an emf induces in the loop which lasts as long as the change in flux continues.

② Second Law - The induced emf in a closed loop or circuit is directly proportional to the rate of change of magnetic flux linked with the closed loop or circuit.

i.e. $\epsilon \propto (-) \frac{d\phi}{dt}$

$$\epsilon = - \left(\frac{d\phi}{dt} \right)$$

* The negative sign is due to Lenz Law.

LENZ LAW -

Current induced in the loop due to changing magnetic flux is such that it tends to oppose the rate of change of magnetic flux.

- Lenz law is in accordance with law of conservation of energy.

INDUCED CURRENT -

If N is the number of turns and R is the resistance of a coil, the magnetic flux linked with its each turn changes by $d\phi$ in short time interval dt , then induced current flowing through the coil is

$$\mathcal{E} = -\frac{d\phi}{dt} \quad I = \frac{|\mathcal{E}|}{R} \Rightarrow I = -\frac{1}{R} \left(\frac{d\phi}{dt} \right) \quad \text{--- (1)}$$

$$\therefore I = \frac{dq}{dt}$$

$$\frac{dq}{dt} = -\frac{1}{R} \left(\frac{d\phi}{dt} \right) \quad \text{using (1)}$$

$$\boxed{q = -\frac{1}{R} \int d\phi}$$

MOTIONAL EMF -

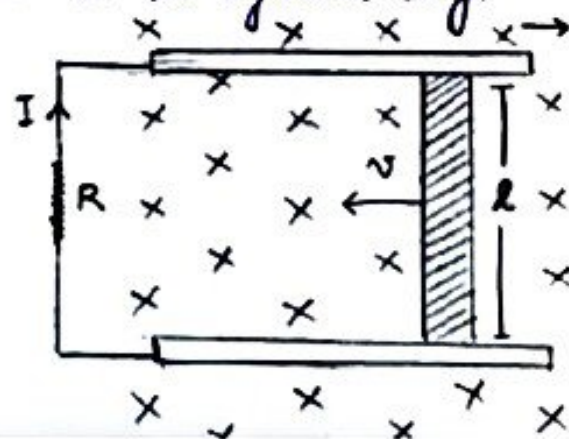
The potential difference induced in a conductor of length l moving with velocity v in a direction perpendicular to magnetic field B is given by

$$\mathcal{E} = \int (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = vBl$$

$B \rightarrow$ magnetic field

$l \rightarrow$ length of conducting wire

$v \rightarrow$ velocity



$$e = -\frac{d\phi}{dt}$$

$$d\phi = BA \quad e = -\frac{d}{dt}(BA)$$

$$A = lx \quad e = -\frac{d}{dt}B(lx)$$

$$e = Blv$$

• emf will not be induced if any two of Blv are parallel.

FORCE

$$I = \frac{Blv}{R}$$

$$F = BIl$$

$$F = \frac{B^2 l^2 v}{R}$$

POWER

$$P = F \times v$$

$$P = \frac{B^2 l^2 v^2}{R}$$

EDDY CURRENT

The current induced in bulk piece of conductor when magnetic flux linked with the conductor changes is known as eddy currents.

$$i = \frac{e}{R}$$

Applications -

1. Magnetic Braking
2. Induction furnace
3. Speedometer
4. Electromagnetic damping
5. Energy meter

Disadvantages -

1. Lot of heat energy is produced which damages the core of material.
2. Excessive heating may lead to fire.
3. Reduces the efficiency of the machine.

Ways to minimize -

1. By laminating the core.
2. By making slots on the conductor surface.

INDUCTANCE -

The flux linkage of a closely wound coil is directly proportional to the current I i.e. $\Phi_B \propto I$. The flux linked with the coil having 'N' turns will be $N\Phi_B \propto I$. The constant of proportionality in this relation is called inductance.

SELF INDUCTANCE

The phenomenon of production of induced emf in a coil, when a current pass through it, undergoes a change.

$\therefore$ Total flux linked with coil, $N\Phi \propto I$

$$\boxed{N\Phi = LI}$$

where, Φ = flux linked with each turn and

L = coefficient of self-induction or self inductance

Also, induced emf, $\epsilon = -\frac{d\Phi}{dt} = -L \frac{dI}{dt}$

SI unit - Henry (H)

where, $L = \frac{\epsilon}{\dots}$

Self Inductance of Long Solenoid -

The magnetic field B at any point inside such a solenoid is constant;

$$B = \frac{\mu_0 N I}{l} = \mu_0 n I$$

where,

μ_0 = magnetic permeability

N = total no. of turns

l = length of the solenoid

n = no. of turns per unit length

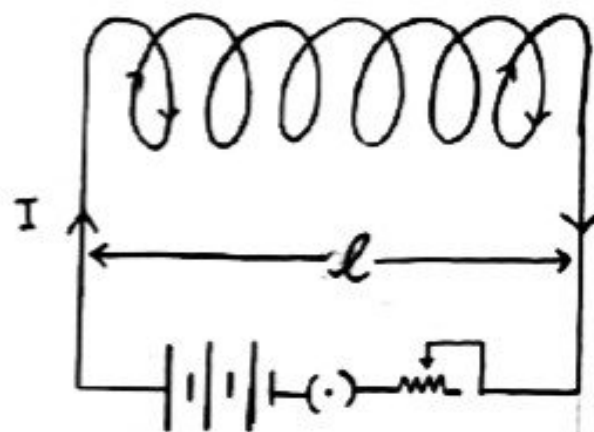
ϕ = $B \times$ area of each turn

$$\phi = \left(\mu_0 \frac{N}{l} I \right) A$$

$$N \phi = L I$$

where $L = \frac{\mu_0 N^2 A}{l}$

$$L = \frac{\mu_0 \mu_r N^2 A}{l}$$



MUTUAL INDUCTANCE -

The phenomenon according to which an opposing emf is produced in a coil as a result of change in current or magnetic flux linked with a neighbouring coil is called inductance.

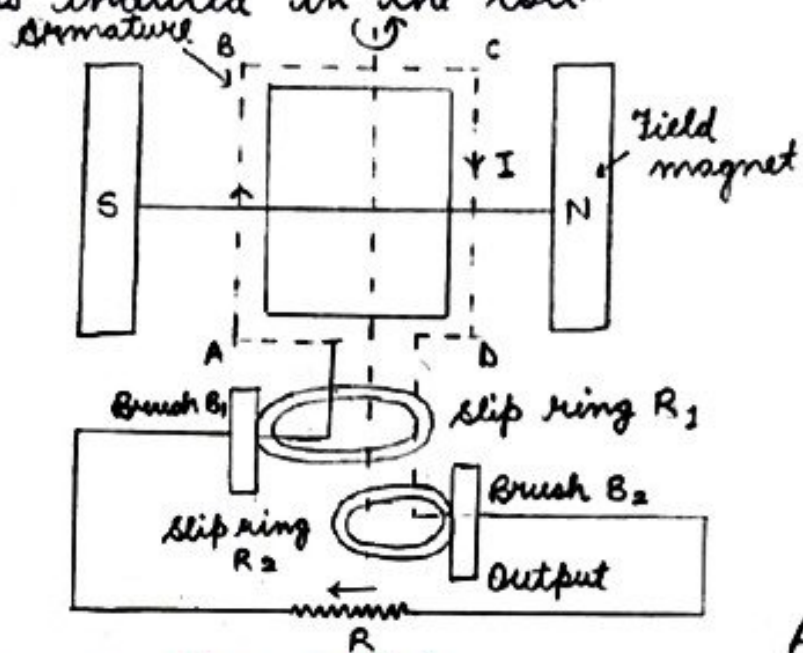
$$\phi \propto I$$

$$\phi = M I$$

$$e = - M \frac{dI}{dt}$$

AC GENERATOR

An AC generator is based on the phenomenon of electromagnetic induction, which states that a coil is rotated in uniform magnetic field, the magnetic flux linked with a conductor changes and an emf is induced in the coil.



AC GENERATOR

Theory and Working

As the armature of coil is rotated in uniform magnetic field, angle θ changes continuously. Therefore, magnetic flux changes and an emf is induced. If e is the emf induced in the coil, then

$$e = - \frac{N d\phi}{dt} \quad \text{or} \quad e = - \frac{d}{dt} (NBA \cos \omega t)$$

$$E = NBA \omega \sin \omega t$$

$N \rightarrow$ no. of turns in the coil

$B \rightarrow$ strength of magnetic field

$A \rightarrow$ area of each turn of coil

$\omega \rightarrow$ angular velocity of rotation of the coil

and $I = \frac{e}{R} = \frac{NBA \omega}{R} \sin \omega t$, $R \rightarrow$ resistance of the coil